LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to aero-elasticity, classification, effects of structural distortion, wing torsional divergence, control effective and reversal
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to aero-elasticity, classification, effects of structural distortion, wing torsional divergence, control effective and reversal. introduction to flutter, control surface flutter, wing bending torsional flutter, factor effecting flutter characteristics, aileron buzz, buffeting, dynamic response, effect of sweep on various critical speeds
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of wing torsional divergence diagrams; Instructor supervision required for safety regarding structural load limits.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define aero-elasticity in one sentence.
(b) List three types of structural distortion.
(c) What is the primary cause of wing torsional divergence?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to aero-elasticity concepts • Classification of aero-elastic effects • Effects of structural distortion • Wing torsional divergence mechanism • Control effectiveness and reversal	L & Ds	PPT, IB	15 min
(b)	• Introduction to flutter phenomena • Control surface flutter characteristics • Wing bending torsional flutter modes • Factors affecting flutter characteristics • Aileron buzz and buffeting effects	L & Ds	PPT, PDF	20 min
(c)	• Dynamic response of airframes • Effect of sweep on critical speeds • Summary of aero-elastic limits • Review of divergence and flutter • Preparation for panel buckling loads	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between divergence and flutter.
(b) How does wing sweep affect critical flutter speed?
(c) Identify two factors that influence flutter characteristics.

12. Home Assignment and Follow Up	Prepare a brief note on the relationship between wing sweep and critical speeds for the next lesson on panel buckling estimation.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Panel bucking, estimation of compressive loads for bucking a panel on simply supported structures. Rigidity due to end conditions on panel, study of heads, to bend panel with regards to variation in end condition shear buckling of plates	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min